

ASSESSMENT 11

Full-Stack AI Developer / Web Developer

Seniority	Mid–Senior
Timebox	4 hours
Assessment	Standalone — no shared inputs or candidate dependencies
Role goal	Build reliable, secure, conversion-aware web products and AI-enabled tools—not no-code-only demonstrations or unreviewed AI-generated code.

Case brief

Build a functional prototype called ‘Beyond X Brief Builder’. A potential client enters company name, sector, objective, audience, needed services, budget range, and desired deadline. The system validates the submission and returns a structured brief summary plus 4–6 discovery questions. The prototype must run without a paid API key using a deterministic mock provider; an optional real LLM provider may be added through environment configuration. Never commit secrets.

Your assignment

Implement the smallest production-minded full-stack prototype that proves frontend, backend/API, validation, AI-provider design, and technical judgment.

Required deliverables

- ☐ Runnable responsive interface with accessible labels, clear errors, loading/success states, and mobile usability.
- ☐ Server/API endpoint with server-side validation and a normalized response schema.
- ☐ AI provider abstraction: deterministic mock works locally; optional LLM integration reads keys from environment variables and handles failure safely.
- ☐ Persist or display the last five local submissions using an appropriate lightweight approach; explain the production data model.
- ☐ At least one automated test covering validation, API behavior, or summary generation.
- ☐ README: setup, architecture, trade-offs, assumptions, security/privacy, performance, analytics events, and production next steps.
- ☐ AI coding log identifying generated code, prompts/tools, manual review, defects found, and changes made.

Recommended time plan

Time	Focus
25 min	architecture and scaffold
145 min	implementation
40 min	testing, security/performance review
30 min	README, AI log, final run

Submission format

Repository or ZIP with source code, README, test instructions/results, screenshots or short demo, and AI coding log. No hard-coded keys. No no-code-only submission.

Assessment rules

- This is a standalone assignment. Use only the information in this brief; do not request, wait for, or use another candidate's output.
- Complete the assessment within the stated timebox. Stop when the time ends and list anything unfinished.
- Use only public, licensed, or synthetic information and assets. Do not upload confidential employer or client material to any AI tool.
- AI use is required, but invisible AI use is not accepted. Include the tools, prompts, outputs used, and the human changes you made.
- State assumptions clearly. Quality of judgment matters more than unsupported certainty or fabricated data.
- Submit both the final output and editable/source files where the role requires them.
- The work must be yours. You may use AI and references, but you must be able to explain every decision and reproduce the workflow.

Before you submit

- ☐ Final output is complete, readable, and opens correctly.
- ☐ Assumptions and sources are visible; no data or achievements were invented.
- ☐ AI tools/prompts and human changes are disclosed.
- ☐ Editable/source files are organized and included where required.
- ☐ The total work stayed within the stated timebox.

Submission declaration: I confirm this is my work, I used no confidential material, and my AI-use log is complete.

Candidate name: _____ Date: _____ Time spent: _____